

Siddharth Maurya

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EDUCATION

IIT BOMBAY

MTECH IN COMPUTER SCIENCE

Jul 2023 | Mumbai, Maharashtra
CGPA: 9.53 / 10.0

IIIT KOTTAYAM

BTECH (HONS) IN COMPUTER SCIENCE

May 2020 | Kottayam, Kerala
CGPA: 8.89 / 10.0

PATENTS

Methods and Systems for Formation and Termination of Payment Channel between Distinct Ledgers
US20240378600A1

Method and System for Securely Shrinking Blockchain with On-Chain State Storage
IN202441048672

SKILLS

Android Development • Distributed Systems • Blockchain Protocols • Digital Identity (OpenID4VC)

Languages:

C++ • C • JavaScript • Kotlin • Swift • Python • Java

ACHIEVEMENTS

GATE CS — 2021

Secured **All India Rank 42** among 100,000+ candidates

IIT Bombay — 2023

Dr. George B Fernandes Award for **Excellence in Research**

Samsung R&D — 2025

Spot Award for D2D Integration for Private Share

LANGUAGES

English (Professional)
Hindi (Native)
Gujarati (Native)

EXPERIENCE

SAMSUNG R&D INSTITUTE | BANGALORE, KARNATAKA

Lead Software Engineer | Mar 2025 — Present

- Led integration of Device-to-Device (D2D) communication for Private Share, achieving **2× faster transfer** speeds and reducing dependency on cloud-based delivery (FCM)
- Implemented a Push Interceptor module, **reducing integration overhead** for seamless integration of Quick Share D2D into existing Private Share workflows
- Owned end-to-end development of a proof-of-concept for **Aadhaar Verifiable Credentials (VC) integration into Samsung Wallet** using OpenID4VCI, in collaboration with UIDAI
- Developed proof-of-concept for **C2PA-based content authenticity verification** within Image Viewer for Private Share, enabling tamper detection for shared media
- Designed mGrid, a **decentralized mobile compute framework** using lightweight blockchain and a custom exchange protocol for off-chain D2D task sharing, ensuring atomic execution and incentive allocation

Senior Software Engineer | Aug 2023 — Feb 2025

- Developed a proof-of-concept iOS application by **integrating an internal cross-platform C++ Private Share SDK**, enabling private file sharing across iOS and Android
- Automated Private Share AAR build and integration pipeline with Quick Share, significantly reducing manual steps to **improve development efficiency**
- Designed **Regenesis protocol for on-chain state storage**, reducing distributed ledger size by 99.7%; published at IEEE UCC 2024
- Optimized a synthetic data generation pipeline for screenshots, applying image processing techniques for text refinement and **improving generation throughput by 50×** via multi-core CPU parallelization

PROJECTS

XPC: Facilitating Interoperability and Scalability for Cross-Chain and Cross-Shard Transactions with Payment Channels

- Worked with **Prof. Vinay Ribeiro** and **Prof. Umesh Bellur** to develop XPC, a system for enabling cross-chain payment channels between independent blockchains; work published at IEEE ICDCS 2025
- Implemented the protocol from scratch, enabling cross-chain payment channel interactions using **MetaMask** and **WebRTC-based** peer-to-peer communication

PUBLICATIONS

- [1] S. Maurya, N. Awathare, V. J. Ribeiro, and U. Bellur. Tombolo: Towards a decentralized interconnected blockchain ecosystem using cross-chain payment channels. In *2025 IEEE 45th International Conference on Distributed Computing Systems (ICDCS)*, pages 1044–1054. IEEE, 2025.
- [2] S. Maurya, V. K. Patra, M. Sinha, P. Kumar, and G. K. Addanki. Scalable regenesis: Towards shrinking blockchain with on-chain state. In *2024 IEEE/ACM 17th International Conference on Utility and Cloud Computing (UCC)*, pages 94–102. IEEE, 2024.